The Impact of Pretraining Task and Data Domains on Transfer Learning for Seismic Facies Segmentation

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***Abstract*—Manual annotation of seismic data for facies segmentation is labor-intensive and costly, leading to limited labeled datasets for training deep learning models. Pretraining strategies, either self-supervised or off-domain, offer a way to leverage large unlabeled or external datasets. This study investigates how pretraining tasks and data domains affect seismic facies segmentation performance. Using Bootstrap Your Own Latent (BYOL) for self-supervised pretraining and ResNet-50-DeepLab V3 as the segmentation model, we evaluate multiple pretraining configurations across three seismic datasets and publicly available pretrained weights. Results show that all pretrained models outperform training from scratch, especially in low-data regimes. Linear readout experiments reveal that self-supervised pretraining on in-domain seismic data yields more specialized and effective representations, whereas full fine-tuning favors models pretrained on large natural image datasets like COCO and ImageNet. These findings suggest that broad visual pretraining provides a robust, adaptable foundation for seismic facies segmentation.**

***Index Terms*—Self-supervised learning, seismic interpretation, BYOL, transfer learning.**

I. Introduction

The analysis of seismic data is crucial for subsurface investigations, such as hydrocarbon exploration. By interpreting facies structures in seismic images, geologists can identify potential reservoirs. However, the manual annotation required for these segmentation tasks is time-consuming and costly, limiting the availability of labeled data for training deep neural networks [1]. While architectures like Convolutional Neural Networks (CNNs) [2] and transformers [3] have shown promise, their complexity demands large datasets to generalize well and avoid overfitting.

A common strategy to mitigate data scarcity is pretraining, where a model is first trained on a pretext task or a different data domain before being fine-tuned on the target task. As illustrated in Figure 1, the model's initial layers (the backbone) learn general features in a pretext task (left), and a new prediction head is then attached for the downstream task (right), such as seismic facies segmentation. Two main pretraining approaches exist: (*i*) supervised learning on a large, labeled dataset from a different domain (*e.g.*, ImageNet), and (*ii*) Self-Supervised Learning (SSL), which uses pretext tasks on unlabeled, in-domain data.

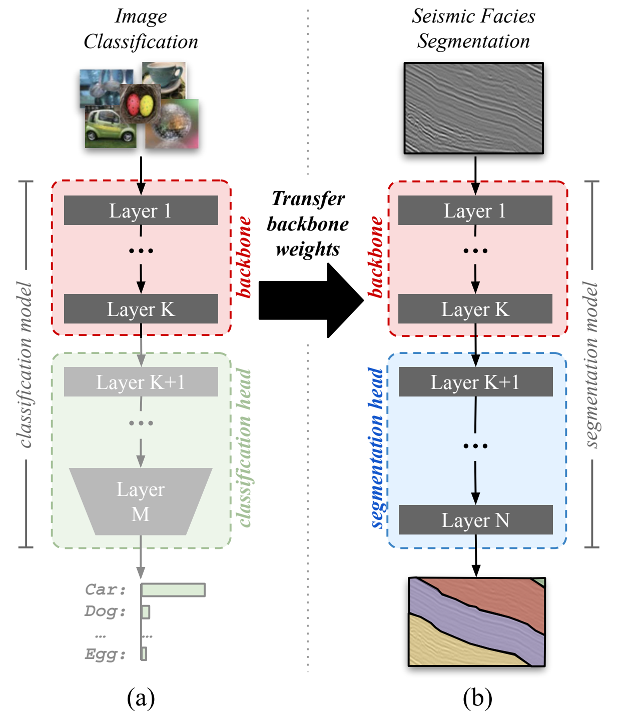

Fig. 1. Example of transfer learning: (a) A backbone is first pretrained on a classification task with natural images; and (b) then refined for a seismic segmentation task.

The relative benefits of SSL for pretraining on in-domain seismic data versus transfer learning from off-domain datasets remain underexplored in seismic interpretation. Seismic data differ from natural images in texture, scale, and statistical structure, raising questions about the generalization of cross-domain representations to geophysical tasks. Conversely, self-supervised models trained on seismic data may learn domain-specific features but lack the diversity and robustness provided by large natural image datasets. While the latter approach, commonly implemented by initializing backbones with publicly available pretrained weights (*e.g.*, ResNet-50 trained on ImageNet), has become standard practice, self-supervised learning leverages in-domain, unlabeled data through pretext tasks that bypass manual annotation.

The primary aim of this work is to investigate the influence of data domains and pretext tasks on the performance of seismic facies segmentation. We question whether SSL pretraining on seismic data offer advantages over using backbones pretrained on natural images or training from scratch. For this

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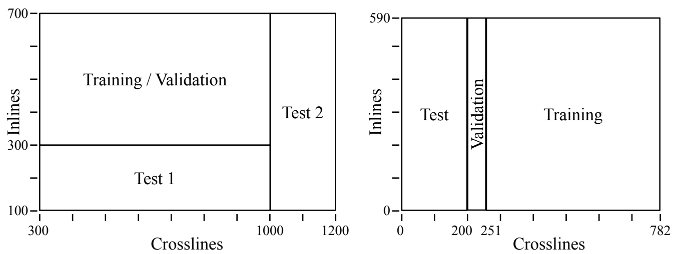


(a) F3 Block (b) Parihaka Block

Fig. 2. F3 and Parihaka datasets partitioning.

TABLE I

PRETRAINING CONFIGURATIONS. ONLY TRAINING AND VALIDATION PARTITIONS FROM F3 AND PARIHAKA WERE USED IN PRETRAINING.

Pretrain Setup	Task	Dataset	Data domain
Cls-ImageNet	Classification	ImageNet	Natural Img.
Seg-COCO	Segmentation	COCO	Natural Img.
BYOL-F3	BYOL	F3	Seismic
BYOL-Par.	BYOL	Parihaka	Seismic
BYOL-F3+Par.	BYOL	F3+Parihaka	Seismic
BYOL-A700	BYOL	A700	Seismic
Scratch	–	–	–

study, we adopt the DeepLab V3 segmentation model, which is widely used for semantic segmentation [4], [5], [6], training it to segment two public seismic datasets: F3 and Parihaka. Our evaluation includes backbones pretrained using five datasets, three with seismic data and two with natural images, and three pretext tasks (BYOL, image classification, and image segmentation) against a baseline trained from scratch.

II. METHODOLOGY

The DeepLab V3 segmentation model uses a ResNet-50 backbone, excluding its final two layers, and employs Atrous Spatial Pyramid Pooling (ASPP) [7] as its segmentation head. To assess the impact of pretraining tasks and data domains on seismic segmentation, we first pretrain the backbone (ResNet-50) under various configurations. We then integrate it into a DeepLab V3 architecture [8] and fine-tune it for the seismic segmentation task.

The following sections present the datasets and tasks considered in this study, as well as the methodologies for pretraining and fine-tuning.

A. Datasets

We used two public, Z-score normalized seismic datasets to assess the performance of the model on the target task: **F3** [9] and **Parihaka** [10]. The F3 dataset partitioning follows Tolstaya *et al.* [2], yielding 992 training, 110 validation, and 400 test images (Fig. 2a). The Parihaka dataset partitioning also follows Tolstaya *et al.*, resulting in 1121 training images, 50 for validation, and 200 test images (Fig. 2b).

For pretraining on seismic data, we employ the F3, Parihaka, and F3+Par (a combination of the two) datasets, together with A700, a private unlabeled seismic dataset comprising over 140,000 seismic crops of size 512×512, extracted from 28 offshore seismic volumes. A700 was constructed from publicly available Brazilian offshore datasets by randomly sampling patches along both the inline and crossline directions. A filtering procedure was applied to remove zero-valued and border patches, ensuring data quality and representativeness. To enhance diversity, approximately 30 patches were sampled per gigabyte of input data, following a sparse and uniformly distributed selection strategy.

For off-domain pretraining, we use the large-scale natural image datasets ImageNet ($\sim$ 1.28 million images) and COCO ($\sim$ 82,000 images filtered by VOC labels) trained on top of ImageNet1K pretrained backbone. Rather than pretraining the ResNet-50 model from scratch on natural image datasets, we adopt the official pretrained weights provided by PyTorch.

B. Pretraining Tasks

The ResNet-50 backbone was pretrained using two approaches: SSL and supervised learning.

1) Self-Supervised Learning (SSL): We adopt Bootstrap Your Own Latent (BYOL) [11], a modern SSL method that learns robust representations by training an *online* network to predict the output of a *target* network processing an augmented view of the same input. The target network's weights are updated as an exponential moving average of the online network's weights. For seismic data, we apply only Random Crop, Horizontal Flip, and slight rotations (±5 degrees) as data augmentations. The online backbone, trained to minimize the Negative Cosine Similarity loss, is subsequently used for the downstream task.

A preliminary study was conducted to determine the optimal BYOL hyperparameters. By evaluating checkpoints using a linear readout protocol [11], [12], we found that an input size of 256×256 and a batch size of 32 provided the best trade-off between performance and computational cost. This configuration was adopted for all subsequent BYOL experiments.

Using this approach, we produce four backbone variants: "BYOL-F3", "BYOL-Par.", "BYOL-F3+Par.", and "BYOL-A700", each trained with the BYOL framework on one of the seismic datasets. Only the training and validation partitions from F3 and Parihaka were used in pretraining.

2) Supervised Learning (Classification and Segmentation): In addition to self-supervised learning, we also employ supervised pretraining using both natural and seismic datasets. The "Seg-COCO" variant corresponds to a backbone pretrained on the COCO dataset for image segmentation, while the "Cls-ImageNet" variant refers to a backbone pretrained on the ImageNet dataset for image classification. For these two datasets, rather than pretraining the backbone from scratch, we utilize the official pretrained weights provided by PyTorch.

Finally, we employ a variant with random weights, identified as "Scratch". Table I summarizes all pretraining configurations.

C. Evaluation of the Quality of Learned Representations

In order to evaluate the quality of representations learned along the pretraining stage, four approaches were analyzed:

- Linear Freeze: employ the linear readout protocol, in which the pretrained backbone is frozen, and only a simple linear head is fine-tuned for the target task.
- Linear Unfreeze: also employ a simple linear prediction head, however, the whole model is fine-tuned.
- DLV3 Freeze: attached the DeepLab V3 prediction head to the pretrained backbone, but fine-tune only the prediction head.
- DLV3 Unfreeze: attached the DeepLab V3 prediction head to the pretrained backbone and fine-tune the whole model.

D. Fine-tuning for Seismic Segmentation

After pretraining, each backbone was integrated with either the DeepLab V3 ASPP segmentation head or a simple linear one and fine-tuned on the F3 or Parihaka normalized datasets to assess backbones performances.

We explored two fine-tuning strategies: **full fine-tuning**, where all model weights are updated, and **freeze**, where only the new segmentation head is trained.

1) Evaluating the models on Low-Data Regime: For the low-data experiments, a data loading strategy was used to ensure balanced and representative samples. Training images were split into inline and crossline pools, from which subsets were selected using a deterministic binary tree sampling method. This recursively selects segment midpoints, distributing samples evenly and reproducibly across each pool. The inline and crossline subsets were then combined to form the final training set.

Using this approach, the seismic segmentation model was fine-tuned with fixed sample sizes: 2, 3, 4, 5, 6, 7, 8, 16, 32, 64, 128, 256, and 512. All nine backbones were evaluated, with each training repeated five times to obtain 95% confidence intervals and reduce stochastic effects.

2) Performance Evaluation Metrics and Computational Infrastructure: We report Accuracy, F_1-score, and mIoU. Experiments ran on NVIDIA Tesla V100 Graphic Processing Units (GPUs). SSL-Pretraining tasks used 4 GPUs, while fine-tuning used a single one.

III. Results and Discussion

This section presents our experimental results, beginning with the representation study conducted along the hyperparameter search (Subsection III-A), followed by results on the quality of learned representations (Subsection III-B) and concluded with low and full-data regime performance (Subsection III-C).

A. Impact of hyperparameters on BYOL pretraining

To identify the optimal BYOL configuration, various input and batch size combinations were tested. Ten checkpoints were saved at fixed intervals during pretraining and evaluated using the *linear readout* protocol on the Parihaka dataset, which emphasizes the quality of the learned representation. Figure 3 shows mIoU versus pretraining steps for three scenarios:

1) Fixed batch size of 8 with input sizes 64, 128, 256, and 512.

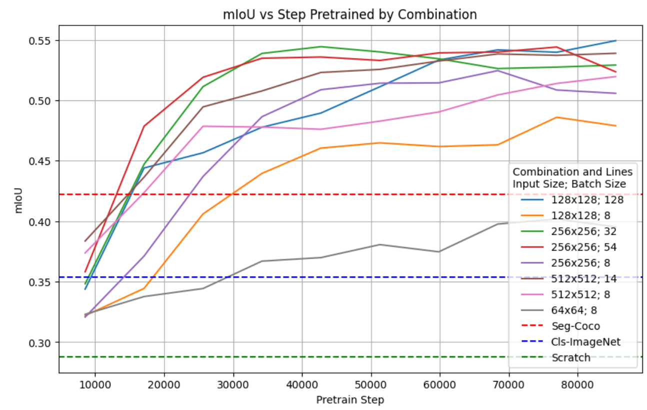


Fig. 3. Performance of different hyperparameter combinations in the linear readout protocol with pretrain and fine-tune on Parihaka.

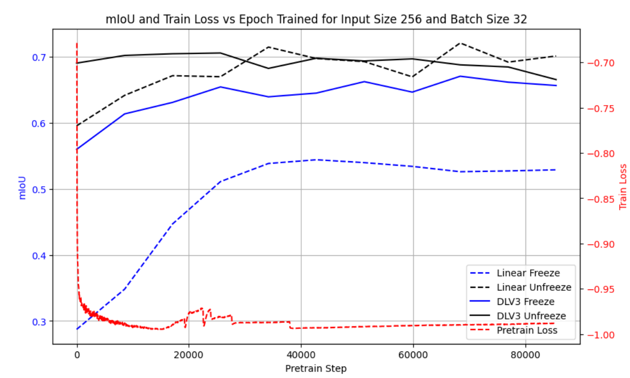


Fig. 4. Analysis of different fine-tuning strategies in Parihaka for the chosen hyperparameter configuration. All training data is used for fine-tuning.

2) Fixed input size of 256×256 with batch sizes 8, 32, and 54.
3) Configurations maximizing GPU memory usage (*e.g.*, 512×512 with batch size 14).

Horizontal lines indicate reference backbones pretrained with supervised methods: Scratch (random initialization), ImageNet, and COCO, where only the linear head was trained.

Figure 3 shows that larger input sizes and batch sizes improve performance, consistent with the BYOL paper [11]. While GPU-maximizing configurations gave the highest performance, 256×256 with batch size 32 provided an excellent trade-off, ranking among the top three in mIoU and converging faster, around 43 000 steps. This motivated its use in all subsequent experiments.

The selected pretext configuration uses the ADAM optimizer (learning rate 10^{-5}, weight decay 10^{-5}), batch size 32, and 256x256 input after Random Crop. Pretraining ran for 85000 steps. Downstream segmentation on F3 and Parihaka used 50 epochs, ADAM (learning rate 10^{-5}), batch size 8, and Cross-Entropy loss. For low-data experiments (less than 100% training data), the number of epochs were adjusted to keep the number of training steps constant.

B. Quality of the learned representations

This section evaluates the quality of learned representations by comparing different fine-tuning strategies (Figure 4).

Figure 4 shows that full fine-tuning using all training data consistently outperforms training with a frozen backbone, for both linear and DeepLab V3 ASPP heads, as more parameters are optimized. With a frozen backbone and the more complex ASPP head, the performance is better than that of a linear head and the results are largely insensitive to pretraining steps, indicating that expressive heads can compensate for limited representation quality given sufficient data.

Under the linear readout protocol, final mIoU depends primarily on the pretrained backbone quality, improving with longer pretraining. The linear head has limited capacity and succeeds only if the backbone representations align well with the task output, making this protocol widely adopted for evaluating representation quality.

TABLE II

MEAN $\pm$ STD MIOU FOR EACH COMBINATION OF TARGET DATASET AND PRETRAIN SETUPS ACROSS DIFFERENT DATA CAPS ON LINEAR READOUT. BEST RESULTS ARE HIGHLIGHTED WITH BOLD FACE AND HIGHEST AVERAGES WITH NO STATISTICAL SEPARATION UNDERLINED. $\star$ INDICATE THE BACKBONE WAS PRETRAINED WITH DATASET USED IN THE DOWNSTREAM TASK.

Target	Pretrain Setup	Amount of training samples on fine-tuning: 2 samples	8 samples	128 samples	100%
F3	BYOL-A700	0.30 ± 0.08	0.43 ± 0.05	0.57 ± 0.04	0.56 ± 0.04
	BYOL-Parihaka	$\underline{\mathbf{0.42 \pm 0.01}}$	0.49 ± 0.01	0.57 ± 0.01	0.57 ± 0.01
	BYOL-F3$\star$	0.38 ± 0.02	0.50 ± 0.02	$\mathbf{0.59 \pm 0.01}$	0.59 ± 0.01
	BYOL-F3+Par.$\star$	0.38 ± 0.03	0.51 ± 0.02	$\mathbf{0.59 \pm 0.01}$	$\mathbf{0.60 \pm 0.01}$
	Seg-COCO	0.27 ± 0.00	0.45 ± 0.00	0.56 ± 0.00	0.56 ± 0.01
	Cls-ImageNet	0.24 ± 0.00	0.36 ± 0.00	0.46 ± 0.00	0.47 ± 0.00
	Scratch	0.36 ± 0.01	0.41 ± 0.01	0.47 ± 0.01	0.47 ± 0.01
Parihaka	BYOL-A700	0.23 ± 0.03	0.38 ± 0.02	0.43 ± 0.02	0.44 ± 0.02
	BYOL-Parihaka$\star$	0.31 ± 0.02	0.41 ± 0.02	0.45 ± 0.02	0.45 ± 0.02
	BYOL-F3	0.28 ± 0.01	0.37 ± 0.01	0.42 ± 0.00	0.43 ± 0.01
	BYOL-F3+Par.$\star$	$\underline{\mathbf{0.33 \pm 0.02}}$	$\underline{\mathbf{0.46 \pm 0.01}}$	$\underline{\mathbf{0.51 \pm 0.01}}$	$\underline{\mathbf{0.51 \pm 0.01}}$
	Seg-COCO	0.23 ± 0.00	0.34 ± 0.00	0.43 ± 0.01	0.43 ± 0.00
	Cls-ImageNet	0.24 ± 0.00	0.30 ± 0.00	0.36 ± 0.00	0.36 ± 0.00
	Scratch	0.21 ± 0.01	0.26 ± 0.01	0.28 ± 0.01	0.28 ± 0.01

Table II applies the linear readout to the F3 and Parihaka datasets across different fine-tuning data regimes (2, 8, 128, and full training samples). Backbones pretrained on the target dataset ($\star$) generally yield representations better aligned with the downstream task than those pretrained on natural image datasets (COCO, ImageNet), highlighting that domain relevance outweighs pretext task choice when features are expected to be linearly related to segmentation outputs.

For F3 downstream tasks, pretraining on Parihaka produces strong representations, whereas the opposite, pretraining on F3 and performing the downstream on Parihaka, is less effective, suggesting Parihaka contains richer features transferable to other datasets.

Finally, BYOL pretraining failed to produce useful representations on the A700 dataset, likely due to lack of data variability or similarity to the target datasets, asking for further investigation.

C. Final Performance in Low-Data and Full-Data Regimes

We now analyze model performance with full fine-tuning. Figures 5a and 5b show DeepLab V3 (backbone + ASPP segmentation head) fine-tuned on F3 and Parihaka across varying number of labeled samples.

In general, models fine-tuned from scratch under-performs relative to other evaluated models, especially in low-data regimes, occasionally overlapping with pretrained models only when more annotated samples are available (*e.g.*, 'BYOL-A700'), highlighting the benefit of pretrained weights in low-data scenarios.

The curves also show that backbones pretrained on large-scale natural image datasets tend to overcome their self-supervised seismic versions as soon as more labeled data is available for fine-tuning. Their better performance in the medium to high data regime suggest that features learned from massive, diverse natural images provide a more robust and adaptable foundation when the network is allowed to update the entire backbone.

TABLE III

MEAN $\pm$ STD PERFORMANCE FOR DIFFERENT COMBINATIONS OF TRAINING AND PRETRAINING DATASETS WITH 100% OF THE DATA.

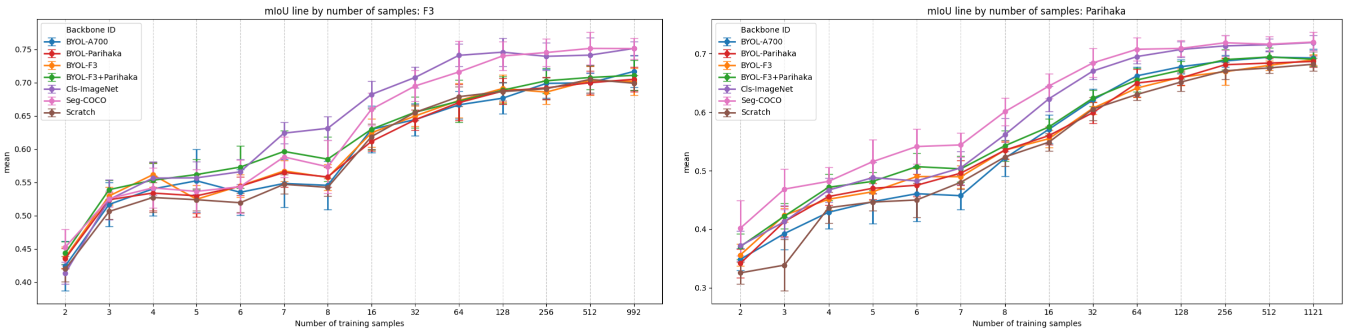
Fine-tune dataset	Pretrain dataset	Mean $\pm$ Std
F3	BYOL-A700	0.72 ± 0.02
	BYOL-Parihaka	0.71 ± 0.02
	BYOL-F3	0.70 ± 0.02
	BYOL-F3+Par.	0.71 ± 0.02
	Seg-COCO	$\underline{\mathbf{0.75 \pm 0.02}}$
	Cls-ImageNet	$\underline{\mathbf{0.75 \pm 0.01}}$
	Scratch	0.70 ± 0.01
Parihaka	BYOL-A700	0.69 ± 0.02
	BYOL-Parihaka	0.69 ± 0.01
	BYOL-F3	0.69 ± 0.01
	BYOL-F3+Par.	0.69 ± 0.01
	Seg-COCO	$\underline{\mathbf{0.72 \pm 0.02}}$
	Cls-ImageNet	$\underline{\mathbf{0.72 \pm 0.01}}$
	Scratch	0.68 ± 0.02

Table III shows the final performance when fine-tuning the model with 100% of the training data, with results averaged across five runs with different random seeds. Backbones pretrained on large-scale natural image datasets achieved the highest mIoU: 'COCO' reached 0.76 on F3, while 'ImageNet' achieved 0.72 on Parihaka.

Wilcoxon tests ($p < 0.05$) confirm the statistical significance of the superior performance of 'COCO' and 'ImageNet' backbones. This indicates that while in-domain pretraining benefits linear readout evaluations, robust features from massive natural image datasets are more advantageous when fine-tuning the entire network. For F3, the best performance is from 'BYOL-A700', followed by 'BYOL-Parihaka' and 'BYOL-F3+Parihaka' for seismic data, just below natural-image models, achieving overall better results. On Parihaka, 'Scratch' is lowest, though only natural-image backbones are statistically superior; all seismic-based backbones show similar means.

IV. CONCLUSION

This work investigates the impact of different data and tasks on the pretraining stage, comparing a modern SSL technique with models pretrained in large out-of-domain datasets. The performance of backbones pretrained with BYOL on different seismic datasets was compared with models pretrained on datasets from natural image domains (ImageNet and COCO) and with a model trained from scratch. The results strongly suggest that using a pretrained backbone is superior to training from scratch, validating the approach to mitigate the scarcity



(a) F3 (b) Parihaka

Fig. 5. Full fine-tuning when varying the amount of training samples on (a) F3 and (b) Parihaka.

of annotated data. Notably, the weights imported from COCO and ImageNet achieved the best final performance after full fine-tuning.

A central observation of the study was the apparent contradiction between the results of the linear readout and full fine-tuning protocols. In the evaluation with the frozen backbone, the representations generated by BYOL on seismic data proved to be more effective and linearly related to the output segmentation mask than those from the scratch and natural image models. However, this advantage was not maintained when fine-tuning the entire network. This indicates that while BYOL learns high-level features more specific to the seismic domain, backbones pretrained on large repositories of natural images have a more robust and rich feature extraction capacity, which, when adapted, allows them to achieve superior performance.

For practical applications, the results suggest that the most effective strategy for seismic facies segmentation remains knowledge transfer from models pretrained on large-scale datasets, such as COCO and ImageNet. This result agrees with previous works [13] that also achieved the same conclusion.

Among the self-supervised approaches, no single pretraining dataset consistently outperformed the others across all evaluations. However, models pretrained on in-domain datasets such as "F3+Par." achieved the best results in the linear readout evaluation, whereas the A700 dataset did not show competitive performance in either the linear readout or the full fine-tuning stages. Even by being the biggest seismic dataset, its volume is not close to the amount of data present in ImageNet.

The poor performance of the backbone pretrained with A700 may be attributed to a combination of limited variety and relatively small size of its collection, which may be insufficient for learning abstract features at the backbone level. Additionally, because this dataset was sampled by gigabyte, larger volumes can overlap with smaller ones, reducing the effective diversity and leading to worse-than-expected performance.

The limitations of this study include the analysis of a single architecture (ResNet-50) and a limited number of public seismic volumes.

The findings of this work raise questions for future research: (*i*) Would it be possible for self-supervised pretraining on an even larger volume of seismic data to outperform knowledge transfer from other domains? (*ii*) Could a hybrid approach, with initial pretraining on natural images followed by a second stage of SSL on seismic data, be advantageous? Future investigations could explore these questions, as well as apply these methods to newer architectures such as Transformers and to three-dimensional (3D) data, which may preserve better structural detail.

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